

CONAN YU

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EDUCATION

University of Waterloo - Bachelor of Applied Science, Computer Engineering	Sept 2026 - June 2031
President's Scholarship of Distinction	
Bayview Secondary School - Ontario Secondary Schools Diploma, IB Diploma	Sept 2021 - June 2026
99.8 Top 6 Average · (7/7): IB Math AA SL, Physics SL, English Lit HL, Chemistry HL, Economics HL (5/5): AP Computer Science A, Calculus BC	

EXPERIENCE

Aquatic Instructor & Lifeguard, City of Richmond Hill	2024 – 2026
Taught swimming to 100+ students in classes of 1–6, tailoring lesson plans and enforcing facility protocols.	
Finance Director, Richmond Hill Youth Action Committee	2023 – 2026
Lead finance team to collect \$1000+ worth of community sponsorships through email campaigns and in-person outreach. Managed committee event documentation and sponsorship allocation.	
Logistics Executive, YR (York Region) Hacks	2023 – 2025
Coordinated event planning and held technical workshops for 500+ participants across two instances of one of Canada's largest hackathons.	

PROJECTS

PawSture

Python · OpenCV · YOLOv8-Pose · Arduino · C++

Real-time posture-correction for students. Body-tracking ML model detects slouching from a webcam and alert users through both hardware(visual) and software(audio+visual) feedback. **1st Place Overall Waterloo Engineering Design Kickstart**. Program uses EMA-smoothing, auto-calibration of posture, a front/side view detection feature with automatic mode/view swapping, and a browser-based interface hosted on local servers. An external Arduino OLED display connected over serial for additional visual alerts when bad posture is sustained.

Denial Therapy

Python · Google ADK · Gemini · Pub/Sub · FastAPI · Google Antigravity

Ambient AI agent that detects health-insurance claim denial events to draft appeal letters, built for Google's 5-day Kaggle AI Agents workshop. Event-driven pipeline on Google Pub/Sub to classify incoming denial events, scrubs PII, and includes human review for drafted appeals. Keyword-based prompt-injection defenses to protect the LLM creating drafts.

Portfolio

HTML · CSS · JavaScript · Canvas API

My portfolio site. Content/details centralized in one .js file for page updates without editing markup. Hash-based routing for navigation, dark theme with 3 canvas-rendered and swappable/toggable animated rhythm-game backgrounds.

AWARDS & CERTIFICATIONS

Hypatia Waterloo Mathematics Contest: Honour Roll	2025
Canadian Open Mathematics Challenge (COMC): Performance with Distinction	2024
Canadian Intermediate Mathematics Contest (CIMC): Honour Roll	2023

SKILLS

Languages: Python, Java, C++, LaTeX. Frameworks & Tools: OpenCV, YOLOv8, Google ADK, Gemini API, FastAPI, Google Pub/Sub, Google Antigravity, SciKit-learn, Pandas, Arduino/C++, Git, GitHub.
Focus: Machine learning, agentic AI, embedded/hardware integration